System Request — AI-Enhanced Client Services Platform

Prepared for the "**Flytics**" Client Services Business Unit

Document purpose

This system request reframes the original Client Services project as an AI-enabled digital platform that lets clients request drone flight services and data analysis through the company website, while improving pilot matching, pricing discipline, operational control, and the value of delivered analytics.

1. Executive Summary

The Client Services Business Unit requires a web-based platform that enables clients to request drone flight services and associated data analysis without owning drone assets directly. The solution will route qualified work to contracted pilots, manage bidding and assignment, and deliver reports and data products to clients through a secure portal.

The proposed enhancement is an AI Services Layer that improves how requests are captured, priced, matched, fulfilled, and reported. This layer reduces manual triage, improves service consistency, increases conversion, and creates higher-margin revenue streams in analytics, subscriptions, and premium turnaround services.

2. Business Context

Project Sponsor	Mahros Mohamed
Business Unit	Client Services
Primary Users	External clients, contracted drone pilots, internal operations staff, finance and support teams
Channel	Company website and secure service portal
Core Outcome	A scalable marketplace for drone flight services and data analysis
Original Conservative Value	$357,500 from new pilot contracts and drone leases plus $565,000 from drone flight services and data analysis

3. Business Need

The business unit has been established to serve clients who need flight services and actionable drone-based analysis, but who do not require direct drone ownership. To support that model, the business needs a digital service platform capable of receiving requests, qualifying demand, distributing opportunities to pilots, selecting the best delivery resource, and returning client-ready outputs promptly and securely.

Without a dedicated system, the new unit faces slower response times, inconsistent scoping, operational friction during pilot assignment, and limited ability to scale analytics-driven offerings.

4. Proposed Solution

Deploy an AI-enhanced Client Services Platform with five integrated capability areas:

- Client request intake through a guided web experience and secure portal.
- Pilot marketplace workflow for bid distribution, ranking, assignment, and scheduling.
- AI Services Layer for request parsing, service recommendation, pricing guidance, pilot matching, risk checks, and report generation.
- Data processing and analytics workflow for imagery upload, analysis, deliverable packaging, and client download.
- Business operations controls for compliance, invoicing, notifications, SLA tracking, and audit history.

5. AI Enhancement Layer

The AI layer should be introduced as a reusable platform service rather than a single feature. Recommended capabilities are listed below.

AI capability	Primary function	Business value
Smart intake	Convert plain-language client requests into structured job requirements.	Fewer incomplete requests; better first-pass scoping
Service recommendation	Suggest the most suitable package, deliverables, and optional add-ons.	Higher conversion and upsell
Pilot matching	Rank pilots by proximity, certification, equipment, availability, performance, and cost fit.	Faster assignment and higher service quality
Pricing guidance	Recommend price ranges from job complexity, location, history, and urgency.	Better margin discipline and bid consistency
Risk and feasibility scoring	Check weather, airspace, travel burden, and permit needs before release to market.	Lower operational risk and fewer failed jobs
Automated analytics	Detect anomalies, changes, counts, or trends from captured imagery and sensor data.	Higher-value outputs and analytics revenue
Report generation	Draft client-ready summaries, findings, and recommended next actions.	Shorter turnaround time and stronger client experience

6. High-Level Business Requirements

1. The system shall enable clients to submit service requests through the company website using a guided intake flow or free-text description.
2. The system shall convert submitted requests into structured mission details including location, service objective, schedule window, deliverables, and urgency.
3. The system shall make eligible requests available to qualified contracted pilots during a configurable bidding window.

4. The system shall support AI-assisted pilot ranking based on proximity, capability, certifications, equipment fit, performance history, and estimated cost competitiveness.
5. The system shall provide AI-assisted pricing guidance to internal staff and, where appropriate, to pilots responding to bids.
6. The system shall assess mission feasibility using weather, airspace, safety, travel, and permit considerations before final assignment.
7. The system shall allow internal staff to approve, reject, or manually override AI recommendations for high-impact decisions.
8. The system shall store uploaded imagery and analysis outputs in a secure client portal with role-based access control and full audit history.
9. The system shall support automated or analyst-reviewed image interpretation and report generation for client delivery.
10. The system shall provide notifications, scheduling updates, status tracking, and SLA alerts for clients, pilots, and operations personnel.
11. The system shall manage payment, invoicing, and delivery confirmation workflows for completed jobs.
12. The system shall maintain compliance records for pilot credentials, insurance, waivers, and customer approvals.

7. Recommended Scope

In scope	Client request portal, pilot bidding workflow, assignment and scheduling, AI intake, pricing guidance, pilot ranking, report generation, secure deliverables portal, notifications, invoicing, compliance records
Out of scope for initial release	Autonomous flight control, hardware manufacturing, advanced ERP replacement, custom client machine-learning model training, full predictive maintenance suite

8. Business Value

The original business case identified conservative annual tangible value of $922,500, composed of $357,500 from new pilot contracts and drone leases and $565,000 from drone flight services and data analysis.

The AI-enhanced version of the platform increases revenue potential in three ways: better client conversion through guided scoping, higher attachment of premium analytics and reporting, and recurring subscription-style monitoring services for repeat customers.

Value driver	Expected effect
Improved conversion	More complete requests and faster first response increase win rates
Operational efficiency	Reduced manual triage, fewer mismatched assignments, and less rework
Analytics upsell	Premium AI analysis commands higher price points than flight-only services

Recurring services	Monthly or quarterly monitoring contracts improve revenue predictability
Enterprise retention	Status tracking, document history, and reliable reporting improve repeat business

9. Constraints, Risks, and Controls

Key constraints	Fast launch expectations, compliance requirements, data security, external API dependencies, pilot supply quality, and the need for explainable AI recommendations
Operational risks	Insufficient pilot coverage, inaccurate scoping, weather disruption, unsafe mission requests, and inconsistent report quality
Controls	Human approval thresholds, credential validation, audit logs, explainable ranking rationale, secured storage, role-based access, and monitored AI outputs

10. Success Measures

- Request-to-assignment cycle time reduced by at least 30 percent after stabilization.
- Increase in request completion quality and decrease in manual clarifications.
- Higher pilot response rate within bid windows for qualified jobs.
- Growth in premium analytics attachment rate and recurring-service contracts.
- Client satisfaction gains through faster delivery, clearer reporting, and transparent status tracking.
- Documented compliance and auditability for all accepted and completed missions.

11. Recommended Delivery Phases

Phase	Focus	Representative deliverables
Phase 1	Digital foundation	Client portal, structured request capture, pilot bidding, assignment workflow, notifications, invoicing, and compliance recordkeeping
Phase 2	AI operations layer	Smart intake, service recommendation, pricing guidance, pilot ranking, feasibility scoring, and AI-assisted support
Phase 3	Advanced analytics products	Automated image analysis, premium reports, enterprise dashboards, subscription monitoring, and deeper trend analysis